

Amogh draft

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Name of Author

Bar-Ilan University, Ramat Gan, Israel

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Abstract

In case you want to write an abstract, here is the palce for it

1 Hybrid CRB with Mixed Random–Deterministic Parameters

1.1 Parameter split

Let

$$\underbrace{\boldsymbol{\theta}_r}_{\text{random}} \triangleq \begin{bmatrix} \boldsymbol{\psi} \\ \boldsymbol{\phi} \end{bmatrix}, \quad \underbrace{\boldsymbol{\theta}_d}_{\text{deterministic}} \triangleq \begin{bmatrix} \boldsymbol{\rho} \\ \boldsymbol{\iota} \\ \sigma_w^2 \end{bmatrix},$$

so only the calibration *gains* and *phases* are modelled as random, while the Toeplitz-row parameters and the noise variance remain fixed (unknown) constants.

1.2 Bayesian justification of the block structure

Bayes' rule factors the joint density of data $\mathbf{z}$ and unknowns:

$$p(\boldsymbol{\theta}_r, \boldsymbol{\theta}_d, \mathbf{z}) = p(\mathbf{z} \mid \boldsymbol{\theta}_r, \boldsymbol{\theta}_d) p(\boldsymbol{\theta}_r) p(\boldsymbol{\theta}_d).$$

We assign an *improper uniform* prior to the deterministic part, $p(\boldsymbol{\theta}_d) \propto 1$; its flat log-density thus carries zero curvature and contributes no Fisher information. Taking the negative expected Hessian of $\ln p(\boldsymbol{\theta}_r, \boldsymbol{\theta}_d, \mathbf{z})$ yields two additive pieces:

$$-\mathbb{E}[\nabla_{\boldsymbol{\theta}}^2 \ln p(\boldsymbol{\theta}_r, \boldsymbol{\theta}_d, \mathbf{z})] = \underbrace{\mathbf{J}_{\text{prior}}^{(\boldsymbol{\psi}, \boldsymbol{\phi})}}_{\text{random block}} + \mathbb{E}_{\boldsymbol{\theta}_r}[\mathbf{J}_{\text{classical}}(\boldsymbol{\theta})].$$

Consequently $\boldsymbol{\theta}_d$ inherits its information *solely* from the likelihood term, allowing every classical FIM entry already derived in the Appendix to be reused unchanged.

1.3 Prior distributions and Fisher block

- **Gains** ($m \geq 2$). One-sided Laplace (scale $b = 1$) truncated to $[0, 2]$:

$$p_{\psi}(\psi_m) = \frac{e^{-\psi_m}}{1 - e^{-2}} \mathbb{1}_{[0, 2]}(\psi_m).$$

Its score variance is $J_{\text{prior}}^{(\psi_m)} = 1$, giving

$$\mathbf{J}_{\text{prior}}^{(\boldsymbol{\psi})} = (1, \dots, 1)_{(M-1) \times (M-1)}.$$

- **Phases** ($m \geq 3$). Bounded uniform $\phi_m \sim \mathcal{U}[-\phi_{\max}, \phi_{\max}]$ has zero curvature, hence $\mathbf{J}_{\text{prior}}^{(\boldsymbol{\phi})} = \mathbf{0}$.

1.4 Hybrid Fisher information matrix

Collecting the results,

$$\boxed{\mathbf{J}_{\text{hyb}} = \mathbf{J}_{\text{prior}} + \mathbb{E}_{\boldsymbol{\theta}_r} [\mathbf{J}_{\text{classical}}(\boldsymbol{\theta})]}, \quad \mathbf{J}_{\text{prior}} = \begin{bmatrix} \mathbf{J}_{\text{prior}}^{(\psi)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

The hybrid CRB then asserts

$$\mathbb{Cov} \llbracket \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta} \rrbracket \succeq \mathbf{J}_{\text{hyb}}^{-1}.$$

All likelihood-dependent terms $\mathbf{J}_{\text{classical}}(\boldsymbol{\theta})$ have already been computed. And since we are computing the **BCRB**

$$\mathbb{E}_{\boldsymbol{\theta}_r} [\mathbf{J}_{\text{classical}}(\boldsymbol{\theta})] = \mathbf{J}_{\text{classical}}(\boldsymbol{\theta})$$

References